

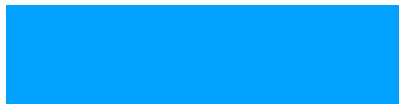
Tokenization

NLP: Fall 2025

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Word structure and subword models

- NLP used to model the vocabulary in simplistic ways based on English
- Tokenize based on spaces into a sequence of "words"
- All novel words at test time were mapped to [UNK] (unknown token)

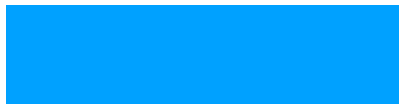
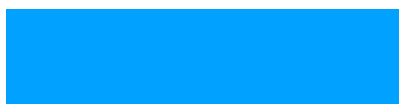
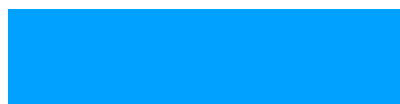
	word	index	embedding
	hat	hat	
	learn	learn	
spell errors	laern	[UNK]	
variations	taaasty	[UNK]	
neologisms	Transformerify	[UNK]	

Byte Pair Encoding algorithm

- Learn a vocabulary of parts of words (subwords)
- Vocabulary of subwords is produced before training a model on the training dataset (larger the better)
- At training and test time the vocabulary is split up into a sequence of known subwords
- Byte Pair Encoding (BPE) algorithm (takes max merges as input)
 - Init subwords with individual characters/bytes and "end of word" token.
 - Using the training data find most common adjacent subwords, merge and add to list of subwords
 - Replace all pairs of characters with new subword token; iterate until max merges

Word structure and subword models

- Common words are kept as part of the vocabulary (ignore morphology)
- Rarer words are split up into subword tokens
- In the worst case, words are split up into characters (or bytes)

	word	index	embedding		
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	learn	learn			
spell errors	laern	la## ##ern			
variations	taaasty	ta## #aa #sty			
neologisms	Transformerify	Transformer## ##ify			